Business Statistics

Measures of Location



Some graphics and datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Measures of Location
Overview

- What is the “location” of a variable?
  - Balance point
- Value-based locations
  - Mean
  - Weighted mean
  - Geometric mean
- Order-based location
  - Median
  - Percentiles
- Frequency-based location
  - Mode

Location of a variable

What is “location?”



Location of a Variable
The "Measure of Central Tendency"

- Recall that a variable takes on multiple values
- A variable's value is only known for a specific observation
- If observations have different values, how can the variable be described with a single representative value?
- That representative value is the "**center**" of a variable's values, and it is called the variable's "**location**"
- There are different values of the center depending on how we consider the data
  - Value-based
  - Order-based
  - Frequency-based

Location of a variable

Value-based locations



Location of a Variable
Location based on the values of the observations

- The **average** is the familiar value-based measure
- Value-based measures use math applied to all observations
- Changing any observation value will change a value-based measure
- Therefore, outliers will “pull” the center towards them
- We will consider:
  - Arithmetic mean (average)
  - Weighted mean
  - Geometric mean

Location of a Variable
The average, or arithmetic mean

- The **mean** is commonly referred to as the average
- As there are other "means", this is technically the "**arithmetic mean**"

$$arithmetic\ mean = \frac{sum\ of\ observations}{count\ of\ observations}$$

- In math notation, assume the are n observations
- Indexed by $i \in \{1, 2, \ldots, n\}$
- The Greek letter sigma $\sum_i$ means to sum over the values x_i indexed by i
- Then the arithmetic mean $\overline{x}$ is

$$\overline{x} = \frac{\sum_i x_i}{n}$$

Location of a Variable
Means for populations and samples

- If collecting all observations from a population is impractical or impossible, a sample is taken from which we draw conclusions about the population
- Distinct notation is used for the values we calculate depending on whether they are from a sample or the population

For the population:

- A measure is a **parameter**
- There are $\boldsymbol{N}$ observations
- The mean is the **parameter** $\boldsymbol{\mu}$

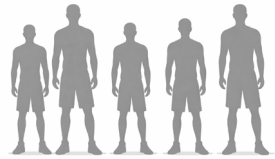
$$\mu = \frac{\sum_i x_i}{N}$$

For a sample:

- A measure is a **statistic**
- There are $\boldsymbol{n}$ observations
- The mean is the **statistic** $\boldsymbol{\overline{x}}$

$$\overline{x} = \frac{\sum_i x_i}{n}$$

Location of a Variable
Arithmetic mean: basketball team height



- A basketball coach is interested in team metrics
- What is their average height?

Basketball Team Player Heights

Observation	Player	Position	X = Height (in)
1	G1	Guard	71.5
2	G2	Guard	73.2
3	F1	Forward	77.8
4	F2	Forward	79.5
5	C	Center	83.7

- There are $N = 5$ observations, indexed by $i \in \{1, 2, 3, 4, 5\}$
- Their heights are $x_i = \{71.5, 73.2, 77.8, 79.5, 83.7\}$
- The arithmetic mean for the population is:

$$\mu = \frac{\sum_i x_i}{N}$$

$$\mu = \frac{71.5 + 73.2 + 77.8 + 79.5 + 83.7}{5}$$

$$\boxed{\mu = 77.1}$$

Location of a Variable
Weighting observations for their importance

- Each **observation** x_i has a weight w_i for its importance
- Each weight w_i is multiplied by its corresponding observation x_i
- To be valid, the weights **MUST** follow 2 rules:
  - Each w_i must be between 0 and 1: $0 \leq w_i \leq 1$
  - The weights must sum to 1: $\sum_i w_i = 1$
- The weighted mean accounts for observations' importance :

$$\overline{x} = \sum_i w_i x_i$$

Location of a Variable
Why isn't the weighted mean divided by n?

- Consider a sample of 3 months of sales:

Monthly Sales		
i = obs	Month	X = Sales ($)
1	Oct	90
2	Nov	100
3	Dec	110

- The arithmetic mean is:

$$\bar{x} = \frac{\sum_i x_i}{n} = \frac{90 + 100 + 110}{3} = \frac{300}{3} = 100$$

$$\bar{x} = \frac{\sum_i x_i}{n} = \frac{90}{3} + \frac{100}{3} + \frac{110}{3} = 30 + 33.3 + 36.7 = 100$$

- Assume each observation has equal importance, $w_i = \frac{1}{3} = 0.33$

$$\bar{x} = \sum_i w_i x_i = (0.33)90 + (0.33)100 + (0.33)110$$

$$\bar{x} = \sum_i w_i x_i = 30 + 33.3 + 36.7 = 100$$

Location of a Variable
Last month's sales are more important

- Consider the sales $X_i = \{90, 100, 110\}$
- More recent data is more important
- Set the weights to reflect this:

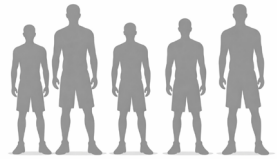
Monthly Sales			
i = obs	Month	**X** = Sales ($)	**w** = weight
1	Oct	90.0	0.2
2	Nov	100.0	0.3
3	Dec	110.0	0.5

$$w_i = \{0.2, 0.3, 0.5\}$$

$$\overline{X} = \sum_i w_i X_i = (0.2)90 + (0.3)100 + (0.5)110$$

$$\overline{X} = \sum_i w_i X_i = 18 + 30 + 55 = 103$$

Location of a Variable
Weighted mean: basketball team height



- A gambler believes tall forwards are the key to a good offense
- She wants the forwards' heights to count for more in the average height?

Basketball Team Player Heights				
Player	Position	X = Height (in)	Weighting	w = weight
G1	Guard	71.5	1.00	0.17
G2	Guard	73.2	1.00	0.17
F1	Forward	77.8	1.50	0.25
F2	Forward	79.5	1.50	0.25
C	Center	83.7	1.00	0.17

- Heights $x_i = \{71.5, 73.2, 77.8, 79.5, 83.7\}$
- Weights $w_i = \{0.17, 0.17, 0.25, 0.25, 0.17\}$
- Weighted mean:

$$\mu = \sum_i w_i x_i = (0.17)71.5 + (0.17)73.2 + (0.25)77.8 + (0.25)79.5 + (0.17)83.7$$

$$\boxed{\mu = 77.4}$$

Location of a Variable
Geometric mean: average rate of change over time

- **Geometric mean**: n^{th} root of the product of n values

$$\overline{X}_G = (X_1 * X_2 * \cdots * X_n)^{1/n}$$

- It is commonly used in finance.
- Consider an investment over time, that results in **different** rates of return in each period
- The **geometric mean** is the **constant** rate of return that would give the same the same change over all of the periods with varying returns
- **Geometric mean rate of return**

$$\overline{R}_G = ((1 + R_1) * (1 + R_2) * \cdots * (1 + R_n))^{1/n}$$

Location of a Variable
Average rate of return

- Consider stock XYZ's price at the end of the month
- The arithmetic mean of the returns is 0
- Geometric mean rate of return factor

Stock XYZ monthly price time-series data

Period	Month	Price	PriceChange	Return	GrowthFactor
0	Jan	$ 100.00	-	-	
1	Feb	$ 110.00	$ 10.00	0.10	1.10
2	Mar	$ 99.00	$ (11.00)	-0.10	0.90
3	Apr	$ 108.90	$ 9.90	0.10	1.10
4	May	$ 98.01	$ (10.89)	-0.10	0.90
5	Jun	$ 107.81	$ 9.80	0.10	1.10
6	Jul	$ 97.03	$ (10.78)	-0.10	0.90
7	Aug	$ 106.73	$ 9.70	0.10	1.10
8	Sep	$ 96.06	$ (10.67)	-0.10	0.90
9	Oct	$ 105.67	$ 9.61	0.10	1.10
10	Nov	$ 95.10	$ (10.57)	-0.10	0.90

$$\overline{R}_G = ((1.10) * (0.90) * (1.10) \ldots * (0.90))^{1/10} - 1$$

$$\overline{R}_G = 0.995 - 1$$

$$\overline{R}_G = -0.005 = -0.5\%$$

Location of a variable

Order-based locations



Location of a Variable
Order-based measures of location

- Order-based measures require that observations be sorted by value
- Outliers do not impact order-based measures

Data:	32	34	30	33	31
Sorted:	30	31	32	33	34
i:	1	2	3	4	5

- Order-based measures require working with both:
  - The **position** i of a **sorted** observation value
    - 5 is the sorted position for the observation value 34
  - The **value** of the ith **sorted** observation
    - 32 is the value of the 3rd sorted observation

Location of a Variable
The median is the middle observation

- The **median** is the "middle", or center, value of sorted observations
- With n observations, the median is the value at position $i = \frac{n+1}{2}$
- If n is odd, this works directly as there is a value at the index $i = \frac{n+1}{2}$

30 31 **32** 33 34

$$n = 5 \quad \Rightarrow \quad \frac{n+1}{2} = \frac{6}{2} = 3$$

Median = the value at $i = 3$

Median = **32**

Location of a Variable
The median for an even n

- If n is even, there is no index i at $\frac{n+1}{2}$ and therefore no value

- The median is then the average of the 2 values on either side of $\frac{n+1}{2}$

30 31 32 33 34 35

$$n = 6 \Rightarrow \frac{n+1}{2} = \frac{7}{2} = 3.5$$

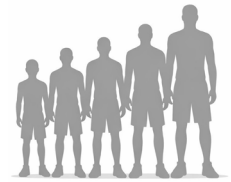
there is no position $i = 3.5$

Take the average of the values at positions $i = 3 \; and \; 4$ *

$$\text{Median} = \frac{32 + 33}{2} = 32.5$$

* These are the $\frac{n}{2}$ and $\frac{n+2}{2}$ positions

Location of a Variable
Median for the basketball team heights



- The team is lined up from shortest to tallest
- What is the height of the player in the middle?
- There are 5 players so $n = 5$
- The median is the observation at the index:

$$\frac{n+1}{2} = \frac{5+1}{2} = \frac{6}{2} = 3$$

- The median is the observation value at $i = 3$ which is 77.8 inches tall

Basketball Team Player Heights

Observation	Player	Position	X = Height (in)
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Location of a Variable
Percentiles are relative positions

- The pth percentile divides a variable's observations such that:
  - Approximately $p\%$ of all observations are below the pth percentile
  - Approximately $(100 - p)\%$ of all observations are above the pth percentile
- With n observations, the pth percentile is the value at position $L = \frac{p}{100}(n + 1)$
- The 50th percentile (median) divides a variable's observations such that:
  - 50% of all observations are below the 50th percentile (the median)
  - $(100 - 50)\% = 50\%$ of all observations are above the 50th percentile (the median)
  - The 50th percentile = value at position $L = \frac{50}{100}(n + 1) = \frac{n+1}{2}$
  - Note that this is the same formula just presented for the median

Location of a Variable
Percentiles are relative positions

- The pth percentile divides a variable's observations into 2 groups:
  - Approximately $p\%$ below and $(100-p)\%$ above the pth percentile
- The 25th percentile = value at position $L = \frac{25}{100}(7+1) = \frac{8}{4} = 2 \implies p_{25} = 55$
- The 75th percentile = value at position $L = \frac{75}{100}(7+1) = \frac{24}{4} = 6 \implies p_{75} = 81$
- The 40th percentile = value at position $L = \frac{40}{100}(7+1) = 3.2$ requires interpolation
- Values at $i = 3, 4$ are 61, 68
- $61 + 0.2 * (68 - 61) = 62.4 \implies p_{40} = 62.4$

Data:	68	94	81	55	42	72	61
Sorted:	42	55	61	68	72	81	94
i:	1	2	3	4	5	6	7
$\boldsymbol{n = 7}$		p_{25}				p_{75}	

Location of a variable

Frequency-based locations



Location of a Variable
Frequency-based measures of location

- The **mode** is the observation that occurs most frequently for the variable

- For these numerical values the mode is 71

62, 68, 71, 71, 71, 74, 77, 81, 85, 90

- For these college majors the mode is Finance

Major
Accounting
Finance
Marketing
Finance
Finance
Management
Economics
Accounting
Finance
Marketing